FACTORIAL-RESIDUE HYPERGRAPHS AND AN AUXILIARY PROBLEM OF ERDŐS

Abstract. We study covering congruences obtained by grouping factorials that have the same residue modulo a prime. For $L \geq 2$, these groups form a weighted edge-colored multihypergraph. We prove that the minimum weight $\kappa(L)$ of a rainbow edge cover is at most $(1-\delta)L\log L$ for some absolute $\delta > 0$ and all sufficiently large L. This minimum is the logarithm of the least modulus arising from the corresponding system of covering congruences. As an application, we prove that there are absolute constants $\eta > 0$ and L_0 such that, for every $L \geq L_0$, at least

$$\frac{((L+1)!)^{\eta}}{100\log L}$$

integers $n \in (2L!, (L+1)!]$ have all prime factors greater than L, with $n-k!$ composite for every $1 \leq k \leq L$. This gives a quantitative solution of an auxiliary problem of Erdős.

1. Introduction

Guy [5, Problem A2] records the following question of Erdős: are there infinitely many primes p such that $p-k!$ is composite whenever $1 \leq k! < p$? The primes 101 and 211 are examples. This question is also listed as Erdős Problem 1059; at the time of writing, the public problem page lists it as open [2].

Alongside Problem 1059, Erdős suggested the following easier auxiliary problem [5, Problem A2]: prove that there are infinitely many pairs (L, n) with

$$L! < n \leq (L+1)!,$$

such that every prime factor of n exceeds L and $n-k!$ is composite for every $1 \leq k \leq L$.

We prove this auxiliary problem in a quantitative form. For an integer $n > 1$, let $P^-(n)$ and $P^+(n)$ denote its least and greatest prime factors, respectively. We call n *L-rough* if $P^-(n) > L$. This condition excludes small prime divisors but allows n to be composite.

The main idea is to use one congruence to make several factorial differences composite. If several factorials have the same residue modulo a prime, requiring n to have that residue makes each of the corresponding differences divisible by the prime. We record these groups in a weighted edge-colored multihypergraph on $\{2, \dots, L\}$. An edge has color q and weight $\log q$ when its factorials share a nonzero residue modulo q, where $L < q < L!$. Let $\kappa(L)$ be the minimum weight of a rainbow edge cover: the selected edges cover all vertices and use each prime color at most once. Section 2 gives the precise definitions and explains the restrictions on q.

Theorem 1.1. *There are absolute constants $\delta > 0$ and L_0 such that*

$$\kappa(L) \leq (1-\delta)L\log L$$

for every integer $L \geq L_0$.

The weight of a cover is the logarithm of the product of its primes, which is the modulus in the Chinese remainder theorem. Thus $\kappa(L)$ is the logarithm of the least modulus obtainable

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from these covering congruences. Theorem 1.1 saves a fixed proportion of the leading cost of covering the indices separately. This saving gives the following result.

Theorem 1.2. *There are absolute constants $\eta > 0$ and L_1 such that, for every integer $L \geq L_1$, if $X = (L+1)!$, then at least*

$$\frac{X^{\eta}}{100 \log L}$$

integers $n \in (2L!, X]$ satisfy $P^-(n) > L$ and have $n - k!$ composite for every $1 \leq k \leq L$.

In particular, solutions of the auxiliary problem exist for every sufficiently large L. They lie in the smaller interval $(2L!, (L+1)!]$. The theorem does not say that any of these integers is prime. The original prime problem is not addressed here.

Factorial residues modulo a fixed prime have been studied from the viewpoints of value sets and distribution; see, for example, [1, 3, 4, 8, 6]. A public AI-assisted working report gives a related congruence-cover formulation for the original prime problem and explains why a fixed modulus cannot settle it [10]; it does not prove the auxiliary theorem established here. Here we use several primes to cover the indices. We first find many pairs of factorials that agree modulo suitable primes, then select pairs with distinct primes and no repeated index.

To prove Theorem 1.1, we use factorials two indices apart. The basic identity is

$$(i+2)! - i! = i!\big(i^2 + 3i + 1\big). \tag{1}$$

The same quadratic, together with the observation that its roots modulo primes are governed by its discriminant 5, appeared earlier in a Math StackExchange discussion of $\gcd(n!+1, (n+2)!+1)$ [7]. Here it is used to produce many low-weight factorial collisions and a uniform sieve construction. If a prime $q > i+2$ divides $i^2 + 3i + 1$, then one fiber modulo q contains both i and $i+2$. A product argument and a one-dimensional upper-bound sieve give linearly many candidate fibers with

$$L < q \ll L^{2-\alpha}$$

for a fixed $\alpha > 0$. An independent-set argument selects a constant proportion having distinct colors and disjoint designated pairs. The remaining vertices are covered separately by primes of size $O(L \log L)$. The paired fibers save a fixed multiple of $L \log L$, which proves Theorem 1.1.

We then count integers in the progression given by the Chinese remainder theorem. A rainbow edge cover with modulus at most $((L+1)!)^{1-\eta}$ leaves enough integers in this progression to exclude all small prime divisors. Counting reduced residue classes modulo the primorial $\prod_{p \leq L} p$ yields the L-rough integers. The restrictions $q < L!$ and $n > 2L!$ make every selected divisor of $n - k!$ proper, and the case $k = 1$ follows from parity. This gives Theorem 1.2.

Section 2 relates the covers to congruences. Sections 3 and 4 provide the estimates for the prime divisors used in the construction, which is carried out in Section 5. Section 6 counts the resulting rough integers.

We write $\pi(x)$ for the number of primes at most x, $\omega(m)$ for the number of distinct prime divisors of m, $\tau(m)$ for its number of positive divisors, and $\varphi(m)$ for Euler's totient function. All logarithms are natural.

2. The factorial-residue hypergraph

Let $L \geq 2$ and put

$$V_L = \{2, 3, \ldots, L\}.$$

Definition 2.1. For a prime $L < q < L!$ and a nonzero residue $a \pmod q$, let $e_{q,a}$ be a labeled hyperedge with incident vertex set

$$E_{q,a} = \{k \in V_L : k! \equiv a \pmod q\}.$$

After empty edges are omitted, these labeled edges form a weighted edge-colored multihypergraph $\mathcal{H}_L$. The edge $e_{q,a}$ has color q and weight $\log q$.

Since $q > L$, every $k!$ with $k \in V_L$ is nonzero modulo q. Therefore, for each fixed q, the fibers $E_{q,a}$ partition V_L, with empty fibers omitted.

Definition 2.2. A *rainbow edge cover* of $\mathcal{H}_L$ is a finite collection

$$\mathcal{C} = \{e_{q,a_q} : q \in Q\}$$

with distinct primes $L < q < L!$, one residue $a_q \in (\mathbb{Z}/q\mathbb{Z})^\times$ for each $q \in Q$, nonempty fibers E_{q,a_q}, and

$$V_L \subseteq \bigcup_{q \in Q} E_{q,a_q}.$$

Its weight is

$$w(\mathcal{C}) = \sum_{q \in Q} \log q.$$

We define

$$\kappa(L) = \inf_{\mathcal{C}} w(\mathcal{C}),$$

where the infimum is over all rainbow edge covers, and put $\kappa(L) = +\infty$ if none exists.

The rainbow condition is necessary because choosing two different fibers of the same color would impose incompatible residues modulo the same prime. The lower bound on q ensures that none of the factorial residues is zero. The upper bound will ensure that the selected divisor of each factorial difference is proper. We omit the index 1 because the difference $n - 1$ will be composite by parity.

Proposition 2.3 (Collision criterion)**.** *Let $2 \le i < j \le L$ and let $q > L$ be prime. Then*

$$\begin{aligned} i! \equiv j! \pmod q \quad &\iff \quad \prod_{r=i+1}^{j} r \equiv 1 \pmod q \\ &\iff \quad q \mid \left(\prod_{r=i+1}^{j} r - 1 \right). \end{aligned}$$

Proof. We have

$$j! = i! \prod_{r=i+1}^{j} r.$$

Also, $q > L \ge i$, so $q \nmid i!$. We may therefore cancel $i!$ modulo q. The three conditions are now equivalent. □

Covering each index with a separate prime gives the following initial bound.

Proposition 2.4. *For every sufficiently large L,*

$$\kappa(L) \le (L-1)\log(20L\log L) = (1 + o(1))L\log L.$$

In particular, a rainbow edge cover exists, and the infimum in Definition 2.2 is a minimum.

Proof. The prime number theorem gives

$$\pi(20L\log L) - \pi(L) > 2L$$

for all sufficiently large L. Choose distinct primes $q_k \in (L, 20L\log L]$ for $k \in V_L$. We also have $20L\log L < L!$ for large L. The edges $e_{q_k,k!}$ form a rainbow edge cover, and their total weight is at most

$$(L-1)\log(20L\log L).$$

There are only finitely many primes between L and $L!$, and each has only finitely many fibers. Hence there are only finitely many rainbow edge covers. The nonempty finite set of their weights has a minimum. □

We next give the exact connection with the Chinese remainder theorem. For sufficiently large L, let $M_{\min}(L)$ be the least squarefree product M of primes $L < q < L!$ for which there exists $A \in (\mathbb{Z}/M\mathbb{Z})^\times$ satisfying both of the following conditions:

(1) for every prime $q \mid M$, there is a $k \in V_L$ such that $A \equiv k! \pmod q$;
(2) for every $k \in V_L$, there is a prime $q \mid M$ such that $A \equiv k! \pmod q$.

Proposition 2.5 (Equivalence with covering congruences)**.** *For every sufficiently large L,*

$$\kappa(L) = \log M_{\min}(L).$$

More precisely, a rainbow edge cover $\mathcal{C}$ gives a modulus

$$M(\mathcal{C}) = \prod_{q\in Q} q$$

with

$$\log M(\mathcal{C}) = w(\mathcal{C}),$$

and every pair (M, A) in the definition of $M_{\min}(L)$ gives a rainbow edge cover of weight $\log M$.

Proof. Let $\mathcal{C} = \{e_{q,a_q} : q \in Q\}$ be a rainbow edge cover. The primes in Q are distinct, so the Chinese remainder theorem gives a unique residue class $A \pmod{M(\mathcal{C})}$ satisfying

$$A \equiv a_q \pmod q \qquad (q \in Q).$$

Every a_q is nonzero, so $(A, M(\mathcal{C})) = 1$. If $k \in E_{q,a_q}$, then $A \equiv k! \pmod q$. Thus $(M(\mathcal{C}), A)$ satisfies the defining conditions for $M_{\min}(L)$. Also,

$$\log M(\mathcal{C}) = \sum_{q\in Q} \log q = w(\mathcal{C}).$$

Conversely, let (M, A) satisfy the defining conditions for $M_{\min}(L)$. For each prime $q \mid M$, choose the edge $e_{q,A \bmod q}$. The residue is nonzero because $(A, M) = 1$, and the second condition says that these edges cover V_L. They use at most one edge of each color and have total weight

$$\sum_{q\mid M} \log q = \log M.$$

Taking the two minima proves the proposition. □

For the selection argument, let $\mathcal{F}$ be a finite family of triples

$$F = (q, a, B_F), \qquad \varnothing \neq B_F \subseteq E_{q,a}.$$

The set B_F is the *designated subset* of F. Define the auxiliary conflict graph $G_{\mathcal{F}}$ on vertex set $\mathcal{F}$ by joining distinct triples $F = (q, a, B_F)$ and $F' = (q', a', B_{F'})$ whenever $q = q'$ or

$$B_F \cap B_{F'} \neq \varnothing.$$

Proposition 2.6 (Weighted independent-set lemma)**.** *Suppose that the conflict graph of $\mathcal{F}$ has maximum degree at most D. If $w(F) \geq 0$ for every $F \in \mathcal{F}$, then there is a subfamily $\mathcal{J} \subseteq \mathcal{F}$ with distinct prime colors and pairwise disjoint designated subsets such that*

$$\sum_{F \in \mathcal{J}} w(F) \geq \frac{1}{D+1} \sum_{F \in \mathcal{F}} w(F).$$

In the special case $w(F) = 1$ for every F, one may choose $\mathcal{J}$ so that

$$|\mathcal{J}| \geq \frac{|\mathcal{F}|}{D+1}.$$

Proof. A finite graph of maximum degree at most D has a proper coloring with at most $D+1$ colors: order the vertices arbitrarily and color them greedily. The color classes are independent sets and partition $\mathcal{F}$. One of them has weight at least the average of their weights. Taking this color class as $\mathcal{J}$ proves the weighted inequality. Taking unit weights gives the last assertion. Independence gives distinct prime colors and pairwise disjoint designated subsets by the definition of the conflict graph. □

Covering the indices in B_F separately has leading cost $|B_F| \log L$. Using one prime q instead saves a fixed multiple of $\log L$ when $q \leq L^{|B_F|-\varepsilon}$ for some $\varepsilon > 0$. In the quadratic construction $|B_F| = 2$, which explains the range $q \ll L^{2-\alpha}$.

3. Prime divisors of a quadratic

We first estimate the frequency with which $i^2 + 3i + 1$ has a sufficiently large prime divisor. Throughout this section, put

$$f(t) = t^2 + 3t + 1.$$

3.1. **Local root counts.** Let χ_5 be the primitive quadratic character modulo 5, and extend it by setting $\chi_5(5) = 0$. We will use the prime number theorem, the Siegel–Walfisz theorem for the fixed modulus 5, and Mertens's estimates in the forms stated in [9, Theorems 3.4, 8.1 and 12.1]. Let

$$\Theta_{\chi_5}(x) = \sum_{p \leq x} \chi_5(p) \log p.$$

For $a \in (\mathbb{Z}/5\mathbb{Z})^\times$, write

$$\vartheta(x; 5, a) = \sum_{\substack{p \leq x \\ p \equiv a \pmod 5}} \log p.$$

The fixed-modulus Siegel–Walfisz estimate and partial summation give $\vartheta(x; 5, a) = x/4 + O(xe^{-c\sqrt{\log x}})$ for some $c > 0$. Since the sum of $\chi_5(a)$ over these four classes is zero, summing with weights $\chi_5(a)$ gives $\Theta_{\chi_5}(x) \ll xe^{-c\sqrt{\log x}}$. Partial summation therefore gives

$$\begin{aligned} \sum_{p \leq x} \chi_5(p) \frac{\log p}{p-1} &= \frac{\Theta_{\chi_5}(x)}{x-1} + \int_2^x \frac{\Theta_{\chi_5}(t)}{(t-1)^2}\, dt + O(1) \\ &= O(1). \end{aligned} \tag{2}$$

Indeed, the integrand is $O(e^{-c\sqrt{\log t}}/t)$, which is integrable on $[2, \infty)$. The same argument, now using the weight $1/(t \log t)$, gives

$$\sum_{p \leq x} \frac{\chi_5(p)}{p} = \frac{\Theta_{\chi_5}(x)}{x \log x} + \int_2^x \Theta_{\chi_5}(t) \frac{\log t + 1}{t^2 (\log t)^2}\, dt + O(1) = O(1).$$

For $p \neq 5$, we have $\mathbf{1}_{\chi_5(p)=1} = (1+\chi_5(p))/2$. Mertens's estimate now gives

$$\sum_{\substack{p\le x\\ \chi_5(p)=1}} \frac{1}{p} = \frac{1}{2}\log\log x + O(1). \tag{3}$$

The identity

$$(2t+3)^2 = 4f(t)+5 \tag{4}$$

shows that, for $p \neq 2,5$, the number of roots of f modulo p is $1+\left(\frac{5}{p}\right)$. Since $5 \equiv 1 \pmod 4$, quadratic reciprocity gives

$$\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right) = \chi_5(p).$$

Thus,

$$\rho(p) = 1+\chi_5(p) \in \{0,2\}. \tag{5}$$

The derivative is $f'(t) = 2t+3$. If a root of f modulo p were not simple, then $f(t) \equiv f'(t) \equiv 0 \pmod p$, and (4) would give $p \mid 5$. Hence, every root under consideration is simple. Hensel's lemma then gives exactly one lift of each root to every modulus p^a. There is no root modulo 2. Modulo 5 there is one root, but it does not lift to a root modulo 25, because

$$f(1+5t) = 5(1+5t+5t^2). \tag{6}$$

3.2. **Large prime factors.**

Lemma 3.1. *For all sufficiently large N,*

$$\#\left\{1 \le i \le N : P^+(f(i)) > \frac{1}{8} i\log i\right\} \ge \frac{N}{5}.$$

Proof. We first bound the number of indices for which every prime factor of $f(i)$ is at most i. Put

$$E_N = \left\{i : \frac{N}{\log N} < i \le N,\ P^+(f(i)) \le i\right\}, \qquad Q_N = \prod_{i\in E_N} f(i).$$

Every prime divisor of Q_N is at most N. For $p \neq 2,5$, the root count modulo prime powers gives

$$\operatorname{ord}_p\left(\prod_{i\le N} f(i)\right) = \sum_{a\ge 1} \#\{i \le N : p^a \mid f(i)\}.$$

For every $a \ge 1$, there are $\rho(p)$ root classes modulo p^a, and hence

$$\#\{i \le N : p^a \mid f(i)\} \le \frac{N\rho(p)}{p^a} + \rho(p).$$

Also, $p^a \mid f(i)$ is possible only when $p^a \le N^2+3N+1$. Consequently, only $O(\log N/\log p)$ values of a occur, and summing the geometric series gives

$$\operatorname{ord}_p\left(\prod_{i\le N} f(i)\right) \le \frac{N\rho(p)}{p-1} + O\left(\rho(p)\frac{\log N}{\log p}\right).$$

The prime 2 contributes nothing. Equation (6) shows that $5 \mid f(i)$ precisely when $i \equiv 1 \pmod 5$, and that in this case $5^2 \nmid f(i)$. Thus, the contribution of $p = 5$ to the logarithm below is $O(N)$. Moreover,

$$\log N \sum_{p \le N} \rho(p) \ll \log N \, \pi(N) \ll N.$$

Therefore, Mertens's estimate together with (2) gives

$$\begin{aligned} \log Q_N &\le N \sum_{p \le N} \frac{(1 + \chi_5(p)) \log p}{p - 1} + O\left(\log N \sum_{p \le N} \rho(p) \right) + O(N) \\ &= N \log N + O(N). \end{aligned}$$

For the corresponding lower bound, uniformly for $N/\log N < i \le N$, we have

$$\log f(i) = 2 \log N + O(\log \log N).$$

It follows that

$$\log Q_N \ge |E_N| \big(2 \log N - O(\log \log N) \big).$$

Comparing this inequality with the upper estimate for $\log Q_N$, we obtain

$$|E_N| \le \left(\frac{1}{2} + o(1) \right) N.$$

The interval $(N/\log N, N]$ contains $N - o(N)$ integers. Thus, at least $(1/2 - o(1))N$ integers in this interval satisfy $P^+(f(i)) > i$. Put

$$\mathcal{S}_N = \left\{ i : \frac{N}{\log N} < i \le N, \ P^+(f(i)) > i \right\}.$$

We have proved that $|\mathcal{S}_N| \ge (1/2 - o(1))N$. It remains to strengthen the lower bound for these prime factors.

Discard from $\mathcal{S}_N$ the indices for which

$$P^+(f(i)) \le \frac{1}{8} N \log N.$$

For every discarded index, the prime $q = P^+(f(i))$ exceeds i. Hence, the relevant indices lie in $[1, q - 1]$ and are distinct modulo q. Since f has at most two roots modulo q, a fixed prime q can occur for at most two indices. Therefore, the number of discarded indices is at most

$$2\pi \left(\frac{1}{8} N \log N \right) = \left(\frac{1}{4} + o(1) \right) N.$$

We started with at least $(1/2 - o(1))N$ indices and discarded at most $(1/4 + o(1))N$ of them. Thus, at least $(1/4 - o(1))N$ indices remain. For each remaining index,

$$P^+(f(i)) > \frac{1}{8} N \log N \ge \frac{1}{8} i \log i.$$

Since $1/4 - o(1) > 1/5$ for all sufficiently large N, at least $N/5$ indices satisfy the required inequality. This proves the lemma. □

4. A sieve estimate for the complementary factor

Write $f(i) = mq$, where $q > L$ is prime. To save on the covering modulus, we also need an upper bound on q, or equivalently a lower bound on its complementary factor m. We now bound the number of indices for which m is too small.

4.1. **Root classes and local counts.** For $0 < \alpha \le 1/2$, let

$$E_\alpha(L) = \#\left\{1 \le i \le L : f(i) = mq \text{ for some } m \le L^\alpha \text{ and prime } q > L\right\}.$$

Thus, $E_\alpha(L)$ counts the indices for which a prime factor $q > L$ has a complementary factor $m \le L^\alpha$. The next lemma bounds this count uniformly in α.

Lemma 4.1. *There is an absolute constant $C_0 \ge 1$ such that, uniformly for $0 < \alpha \le 1/2$ and all sufficiently large L,*

$$E_\alpha(L) \le C_0\left(\alpha L + \frac{L}{\log L} + L^{(1+\alpha)/2}(\log L)^2\right).$$

The lower threshold for L may be chosen independently of α.

Proof. We begin by counting the roots of f modulo m. For $m \ge 1$, write

$$R(m) = \#\{a \pmod m : f(a) \equiv 0 \pmod m\}.$$

The prime-power values from the previous section are

$$R(2^e) = 0 \quad (e \ge 1), \qquad R(5) = 1, \qquad R(5^e) = 0 \quad (e \ge 2),$$

and, for $p \ne 2, 5$,

$$R(p^e) = 1 + \chi_5(p) \quad (e \ge 1).$$

The Chinese remainder theorem therefore gives

$$R(m) = \begin{cases} 2^{\omega(m/(m,5))}, & 2 \nmid m,\ 25 \nmid m, \text{ and} \\ & \chi_5(p) = 1 \text{ for every } p \mid m,\ p \ne 5, \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

Now fix $m \le L^\alpha$ and a root $a \pmod m$, represented by $0 \le a < m$. Every integer $i \equiv a \pmod m$ can be written as $i = a + mt$. Accordingly, let

$$i = a + mt, \qquad T_{a,m} = \{t \in \mathbb{Z} : 1 \le a + mt \le L\}, \qquad H = |T_{a,m}|.$$

The length of this interval of t values is

$$H = \frac{L}{m} + O(1), \qquad H \asymp \frac{L}{m}, \qquad H \ge L^{1/2} - 1.$$

Since $m \mid f(a)$, we can divide $f(a + mt)$ by m and obtain the integral polynomial

$$G_{a,m}(t) = \frac{f(a+mt)}{m} = mt^2 + (2a+3)t + \frac{f(a)}{m}.$$

This polynomial has discriminant 5. It is also primitive. Indeed, if a prime divided all three coefficients, then its square would divide $(2a+3)^2 - 4f(a) = 5$, which is impossible.

If $\ell \nmid 10m$ is prime, then the affine map $t \mapsto a + mt$ is a bijection modulo ℓ. Consequently,

$$\#\{t \pmod \ell : G_{a,m}(t) \equiv 0 \pmod \ell\} = 1 + \chi_5(\ell).$$

Let

$$\mathcal{P}_0 = \{\ell : \ell \text{ is prime and } \ell \nmid 10\}, \qquad \mathcal{P}_m = \{\ell \in \mathcal{P}_0 : \ell \nmid m\},$$

and extend ρ multiplicatively to squarefree integers supported on $\mathcal{P}_m$, with $\rho(d) = 0$ otherwise. For supported squarefree d, counting the root classes inside $T_{a,m}$ gives

$$\#\{t \in T_{a,m} : d \mid G_{a,m}(t)\} = H\frac{\rho(d)}{d} + r_d, \qquad |r_d| \le \rho(d). \tag{8}$$

4.2. **Application of the fundamental lemma.** We check that the constants in the fundamental lemma can be chosen independently of m and a. Define the nonnegative weights

$$a_n = \#\{t \in T_{a,m} : G_{a,m}(t) = n\}, \qquad X_{a,m} = \sum_n a_n = H.$$

Thus, repeated values of $G_{a,m}$ are counted with their multiplicities. For d supported on $\mathcal{P}_m$, equation (8) says

$$\sum_{\substack{n\geq 1\\ d\mid n}} a_n = H\frac{\rho(d)}{d} + r_d.$$

This is precisely Axiom 1 of [9, pp. 185–186], with local density $\nu(d) = \rho(d)$ and remainder r_d. We also have $0 \leq \rho(\ell) < \ell$ for every $\ell \in \mathcal{P}_m$.

Next we need the dimension condition. Mertens's estimate and (2) give, uniformly for $w \geq 2$,

$$\sum_{\substack{\ell\leq w\\ \ell\nmid 10}} \frac{\rho(\ell)\log\ell}{\ell} = \sum_{\ell\leq w}\frac{\log\ell}{\ell} + \sum_{\ell\leq w}\chi_5(\ell)\frac{\log\ell}{\ell} + O(1) = \log w + O(1). \tag{9}$$

Thus, every finite truncation of the base prime set $\mathcal{P}_0$ satisfies Axiom $2'$ of [9, p. 188] with $\kappa = 1$, uniformly in the truncation point. The additional local condition in that axiom holds, for example, with $k = 2$ and $\varepsilon = 1/2$, since $\rho(\ell) \in \{0, 2\}$ and every prime in $\mathcal{P}_0$ for which $\rho(\ell) = 2$ is at least 11. The cited source therefore gives Axiom 2 for every finite truncation of $\mathcal{P}_0$, with the same absolute constant C.

The deletion of primes dividing m requires a uniformity check. We do not claim that Axiom $2'$ remains unchanged after deleting them. If $3/2 \leq y_1 \leq y_2$, then

$$\prod_{\ell\in\mathcal{P}_m\cap(y_1,y_2]}\left(1-\frac{\rho(\ell)}{\ell}\right)^{-1} \leq \prod_{\ell\in\mathcal{P}_0\cap(y_1,y_2]}\left(1-\frac{\rho(\ell)}{\ell}\right)^{-1}.$$

After both sides are restricted to any common finite truncation, the left side is the product that occurs in Axiom 2 for the truncated set $\mathcal{P}_m$. Hence, every such truncation inherits Axiom 2 from the base truncation with the same absolute constants. This proves the required uniformity in m and a.

Set $z = H^{1/20}$ and

$$\mathcal{Q}_{m,z} = \{\ell \in \mathcal{P}_m : \ell \leq z\}, \qquad P_m(z) = \prod_{\ell\in\mathcal{Q}_{m,z}} \ell.$$

For sufficiently large L, we have $z < L$. Therefore, every prime value $G_{a,m}(t) > L$ is counted by the sifted set $\{t \in T_{a,m} : (G_{a,m}(t), P_m(z)) = 1\}$. Let $S_{a,m}$ denote the number of $t \in T_{a,m}$ for which $G_{a,m}(t)$ is a prime exceeding L. If $\mathcal{Q}_{m,z}$ is empty, the estimate below is immediate. Otherwise, let y be its largest prime. We apply [9, Theorem 18.11(a), p. 190] to the finite prime set $\mathcal{Q}_{m,z}$ with $u = 10$. In the notation of that theorem, its main term is

$$\left(1 + O_{\kappa,C}(u^{-u/2})\right) HV_m(z),$$

and its remainder is bounded by

$$O\left(\sum_{\substack{d\leq y^u\\ d\mid P_m(z)}} |r_d|\right).$$

Since $y \le z$, we have $y^u \le z^{10} = H^{1/2}$. Taking an upper bound and enlarging the remainder sum, we obtain

$$S_{a,m} \ll HV_m(z) + \sum_{\substack{d \le H^{1/2} \\ d \mid P_m(z)}} |r_d| \le HV_m(z) + \sum_{d \le H^{1/2}} \rho(d), \tag{10}$$

where

$$V_m(z) = \prod_{\substack{\ell \le z \\ \ell \nmid 10m}} \left(1 - \frac{\rho(\ell)}{\ell}\right).$$

To bound the remainder, we use $\rho(d) \le \tau(d)$. Hence,

$$\sum_{d \le H^{1/2}} \rho(d) \le \sum_{ab \le H^{1/2}} 1 \ll H^{1/2} \log H.$$

It remains to estimate the sieve product. First put

$$V_0(z) = \prod_{\substack{\ell \le z \\ \ell \nmid 10}} \left(1 - \frac{\rho(\ell)}{\ell}\right).$$

By (3) and the convergence of $\sum_p p^{-2}$, we have

$$\log V_0(z) = \sum_{\substack{\ell \le z \\ \chi_5(\ell)=1}} \log\left(1 - \frac{2}{\ell}\right) = -2 \sum_{\substack{\ell \le z \\ \chi_5(\ell)=1}} \frac{1}{\ell} + O(1) = -\log\log z + O(1).$$

The primes dividing m were removed from this product. To measure the cost of removing them, define

$$h(m) = \prod_{\substack{p \mid m,\ p \ne 5 \\ \chi_5(p)=1}} \left(1 - \frac{2}{p}\right)^{-1}.$$

Then, omitting from $V_0(z)$ the primes that divide m gives

$$V_m(z) = V_0(z) \prod_{\substack{\ell \mid m,\ \ell \le z \\ \ell \ne 5,\ \chi_5(\ell)=1}} \left(1 - \frac{2}{\ell}\right)^{-1} \ll \frac{h(m)}{\log z} \ll \frac{h(m)}{\log L}$$

uniformly for $m \le L^\alpha$. The last comparison follows from $H \ge L^{1/2} - 1$, which gives $\log z \asymp \log L$. Substituting this into (10), we obtain

$$S_{a,m} \ll \frac{Hh(m)}{\log L} + H^{1/2} \log L. \tag{11}$$

4.3. **Summation over complementary factors.** If i is counted by $E_\alpha(L)$, then for at least one representation $f(i) = mq$, the residue class $a \equiv i \pmod m$ is one of the $R(m)$ roots and $G_{a,m}(t) = q$ for the corresponding t. We sum over all such m and all such roots. This is a union bound, so an index with more than one representation may be counted more than once. Since $m \le L^\alpha \le L^{1/2}$, we also have

$$H \le \frac{L}{m} + 1 \le \frac{2L}{m}.$$

Put $Y := L^{\alpha}$. Equation (11) now gives

$$E_{\alpha}(L) \ll \frac{L}{\log L} \sum_{m \le Y} \frac{R(m)h(m)}{m} + L^{1/2} \log L \sum_{m \le Y} \frac{R(m)}{\sqrt{m}}. \tag{12}$$

The two sums on the right are estimated separately. For the first, define $g(m) = R(m)h(m)$. Equation (7) and the definition of h show that g is nonnegative and multiplicative, and that it vanishes outside the admissible support. If $p \neq 5$ and $\chi_5(p) = 1$, then

$$1 + \sum_{e \ge 1} \frac{g(p^e)}{p^e} = 1 + \frac{2p}{(p-1)(p-2)} = 1 + \frac{2}{p} + O(p^{-2}).$$

The prime 5 contributes the bounded local factor $1 + 1/5$, while inert primes and 2 contribute 1. Therefore, positivity, the Euler product, and (3) give, for $Y \ge 1$,

$$\begin{aligned} \sum_{m \le Y} \frac{R(m)h(m)}{m} &\le \prod_{p \le Y} \left(1 + \sum_{e \ge 1} \frac{g(p^e)}{p^e} \right) \\ &\ll \exp\left(2 \sum_{\substack{p \le Y \\ \chi_5(p)=1}} \frac{1}{p} + O(1) \right) \ll \log(2Y). \end{aligned}$$

For the second sum, we use $R(m) \le \tau(m)$. Thus,

$$\begin{aligned} \sum_{m \le Y} \frac{R(m)}{\sqrt{m}} &\le \sum_{ab \le Y} \frac{1}{\sqrt{ab}} \\ &\ll \sum_{a \le Y} \frac{1}{\sqrt{a}} \sqrt{\frac{Y}{a}} \ll \sqrt{Y} \log(2Y). \end{aligned}$$

Finally,

$$\log(2Y) \le \alpha \log L + O(1),$$

uniformly for $0 < \alpha \le 1/2$. Substituting these bounds into (12) gives

$$E_{\alpha}(L) \ll \alpha L + \frac{L}{\log L} + L^{(1+\alpha)/2} (\log L)^2,$$

with an absolute implied constant and an L-threshold independent of α. This proves the lemma. □

5. Construction of the cover

Lemmas 3.1 and 4.1 provide the required arithmetic input. The following proposition extracts a family of disjoint pairs with distinct associated primes and covers the remaining indices individually.

Proposition 5.1. *Let C_0 be as in Lemma 4.1, and set*

$$\alpha_0 = \frac{1}{1000 C_0}, \qquad \delta_0 = \frac{\alpha_0}{200}, \qquad \eta_0 = \frac{\delta_0}{2}.$$

For every sufficiently large L, there exist a set $I \subseteq \{2, \dots, L-2\}$, distinct primes q_i for $i \in I$, and distinct primes r_k, indexed by

$$U = \{2, \dots, L\} \setminus \bigcup_{i \in I} \{i, i+2\}.$$

Put

$$M = \prod_{i \in I} q_i \prod_{k \in U} r_k.$$

These objects satisfy the following conditions:

(1) *The pairs* $\{i, i+2\}$, $i \in I$, *are pairwise disjoint, and* $|I| \geq L/100$.
(2) *For every* $i \in I$,
$$L < q_i < 2L^{2-\alpha_0}, \qquad q_i \mid f(i).$$
(3) *For every* $k \in U$,
$$L < r_k \leq 20L \log L.$$
(4) *The primes* q_i, $i \in I$, *and* r_k, $k \in U$, *are pairwise distinct.*
(5) *The modulus* M *satisfies*
$$\log M \leq (1 - \delta_0) L \log L, \qquad M \leq ((L+1)!)^{1-\eta_0}.$$

Moreover, every prime divisor of M *is less than* $L!$.

Proof. The choice of α_0 in the proposition and Lemma 4.1 give

$$C_0 \alpha_0 L = \frac{L}{1000}.$$

Since $\alpha_0 < 1$ is fixed, we also have

$$C_0 \left(\frac{L}{\log L} + L^{(1+\alpha_0)/2} (\log L)^2 \right) = o(L).$$

Consequently,

$$E_{\alpha_0}(L) \leq \frac{L}{500}$$

for sufficiently large L. On the other hand, Lemma 3.1 supplies at least $L/5$ indices $i \leq L$ for which

$$q_i := P^+(f(i)) > \frac{1}{8} i \log i.$$

We discard the indices $i < L/10$, the two endpoints $L-1, L$, and those counted by $E_{\alpha_0}(L)$. The number remaining is at least

$$\frac{L}{5} - \frac{L}{10} - \frac{L}{500} - 2 = \frac{49L}{500} - 2 \geq \frac{9L}{100}$$

for sufficiently large L. For every remaining index,

$$q_i > \frac{1}{8} i \log i \geq \frac{L}{80} \log \left(\frac{L}{10} \right) > L$$

once L is sufficiently large. Write $f(i) = m_i q_i$. Then $m_i > L^{\alpha_0}$, since otherwise this index would be counted by $E_{\alpha_0}(L)$. Since $f(i) < 2L^2$ for $i \leq L$, it follows that

$$q_i < 2L^{2-\alpha_0}.$$

The identity (1) now gives

$$i! \equiv (i+2)! \pmod{q_i}.$$

For each remaining index i, let

$$F_i = (q_i, i!, B_i), \qquad B_i = \{i, i+2\}.$$

The identity above shows that $B_i \subseteq E_{q_i, i!}$. We apply Proposition 2.6. Two designated pairs $\{i, i+2\}$ and $\{j, j+2\}$ intersect only when $j = i$ or $j = i \pm 2$. Thus, a fixed vertex of the auxiliary conflict graph has at most two neighbors arising from an overlap of designated pairs.

A fixed prime $q > L$ occurs for at most two starting indices, because all starts are less than q and f has at most two roots modulo q. Hence a fixed vertex has at most one additional neighbor of the same prime color. The conflict graph therefore has maximum degree at most 3. Proposition 2.6, with unit weights, gives at least

$$\frac{1}{4} \cdot \frac{9L}{100} = \frac{9L}{400}$$

vertices whose associated edges have distinct colors and whose designated pairs are pairwise disjoint. For sufficiently large L, we may therefore choose a subfamily indexed by a set I with

$$|I| = \left\lceil \frac{L}{100} \right\rceil.$$

The set of remaining indices has cardinality $|U| = L - 1 - 2|I|$. The prime number theorem gives

$$\pi(20L \log L) - \pi(L) > 2L$$

for sufficiently large L. After excluding the $|I|$ primes already used for pairs, more than $2L - |I| > |U|$ primes remain. Therefore, we may assign a distinct prime $r_k \in (L, 20L \log L]$ to every $k \in U$.

It remains to estimate the product of the selected primes. Put $r = |I|$ and $s = |U| = L-1-2r$. Then

$$\begin{aligned} \log M &\le r\big((2 - \alpha_0) \log L + \log 2\big) + s\big(\log L + \log\log L + \log 20\big) \\ &= (L - 1 - \alpha_0 r) \log L + s \log\log L + O(L) \\ &\le \left(L - 1 - \frac{\alpha_0 L}{100}\right) \log L + L \log\log L + O(L). \end{aligned}$$

For sufficiently large L, the lower order terms satisfy

$$L \log\log L + O(L) \le \frac{\alpha_0}{200} L \log L,$$

and therefore

$$\log M \le \left(1 - \frac{\alpha_0}{200}\right) L \log L.$$

This is the first estimate in the proposition because $\delta_0 = \alpha_0/200$. Since

$$\log((L+1)!) = L \log L - L + O(\log L),$$

we have $\log((L+1)!) \ge L \log L - 2L$ for sufficiently large L. Since $\eta_0 = \delta_0/2$, we obtain

$$\begin{aligned} (1 - \eta_0) \log((L+1)!) - \log M &\ge (1 - \eta_0)(L \log L - 2L) - (1 - \delta_0) L \log L \\ &= (\delta_0 - \eta_0) L \log L - 2(1 - \eta_0) L > 0 \end{aligned}$$

for sufficiently large L. This proves $M \le ((L+1)!)^{1-\eta_0}$. The displayed ranges also show that every selected prime is less than $L!$ once L is sufficiently large. Finally, we take the maximum of the finitely many absolute thresholds imposed above. This completes the construction. □

Proof of Theorem 1.1. Choose the sets and primes from Proposition 5.1. For each $i \in I$, the identity (1) and the divisibility $q_i \mid f(i)$ give

$$i! \equiv (i+2)! \pmod{q_i}.$$

Hence the edge $e_{q_i, i!}$ covers both i and $i+2$. For each $k \in U$, the edge $e_{r_k, k!}$ covers k. All the primes are distinct, so these edges form a rainbow edge cover $\mathcal{C}_L$. Proposition 5.1 gives

$$\kappa(L) \le w(\mathcal{C}_L) = \log M \le (1 - \delta_0) L \log L.$$

Taking $\delta = \delta_0$ proves the theorem. □

6. From rainbow covers to rough integers

Once a cover has been found, the following theorem counts rough integers in the corresponding progression. Only the size of the modulus is needed.

Theorem 6.1 (Transference theorem)**.** *Fix $0 < \eta < 1$. For every sufficiently large L, depending only on η, let $\mathcal{C}$ be a rainbow edge cover of $\mathcal{H}_L$ and put*

$$X = (L+1)!, \qquad M = M(\mathcal{C}).$$

If $M \le X^{1-\eta}$, then at least

$$\frac{X^\eta}{100 \log L}$$

integers $n \in (2L!, X]$ satisfy $P^-(n) > L$ and have $n - k!$ composite for every $1 \le k \le L$.

Proof. Write

$$\mathcal{C} = \{e_{q,a_q} : q \in Q\}.$$

The primes in Q are distinct, so the Chinese remainder theorem gives a unique residue class $A \pmod M$ satisfying

$$A \equiv a_q \pmod q \qquad (q \in Q). \tag{13}$$

Every a_q is nonzero, and hence $(A, M) = 1$.

Let

$$\mathcal{T} = \{t \in \mathbb{Z} : 2L! < A + tM \le X\}, \qquad H = |\mathcal{T}|.$$

The set $\mathcal{T}$ is an interval of H consecutive integers. Its length gives

$$H \ge \frac{X - 2L!}{M} - 1 = \frac{L-1}{L+1}\frac{X}{M} - 1.$$

For $L \ge 3$, the right side is at least $X/(2M) - 1$. Also, $M \le X^{1-\eta}$ gives $X/M \ge X^\eta \to \infty$. Therefore, for all sufficiently large L,

$$H \ge \frac{X}{3M} \ge \frac{X^\eta}{3}. \tag{14}$$

Put

$$W = \prod_{p \le L} p.$$

Every prime factor of M exceeds L, so $(M, W) = 1$. Writing $\vartheta(y) = \sum_{p \le y} \log p$, the prime number theorem gives

$$\log W = \vartheta(L) = L + o(L).$$

Stirling's formula and (14) give

$$\begin{aligned} \log \frac{H}{2W} &\ge \eta \log X - \log 6 - \vartheta(L) \\ &= \eta L \log L - (1 + \eta + o(1))L - \log 6 \longrightarrow +\infty. \end{aligned}$$

Thus $H \ge 2W$ for sufficiently large L.

Multiplication by M is invertible modulo W. Hence the map

$$t \longmapsto A + tM \pmod W$$

is a bijection on $\mathbb{Z}/W\mathbb{Z}$. Every block of W consecutive values of t contains exactly $\varphi(W)$ values for which $(A + tM, W) = 1$. Since $H \ge 2W$, at least

$$\left\lfloor \frac{H}{W} \right\rfloor \varphi(W) \ge \frac{H}{2}\frac{\varphi(W)}{W}$$

values $t \in \mathcal{T}$ have this property. Mertens's product theorem gives

$$\frac{\varphi(W)}{W} = \prod_{p \le L} \left(1 - \frac{1}{p}\right) \ge \frac{1}{4 \log L}$$

for sufficiently large L. It follows that the number of integers $n = A + tM$ in $(2L!, X]$ with $(n, W) = 1$ is at least

$$\frac{H}{8 \log L} \ge \frac{X}{24M \log L} \ge \frac{X^{\eta}}{24 \log L} \ge \frac{X^{\eta}}{100 \log L}.$$

Every such n is L-rough.

It remains to check the factorial differences. Let $2 \le k \le L$. Since $\mathcal{C}$ is a rainbow edge cover, there is a $q \in Q$ such that $k \in E_{q,a_q}$. Equations (13) and the definition of the fiber give

$$q \mid n - k!.$$

This divisor is proper. Indeed, $q < L!$ by Definition 2.1, and

$$n - k! > 2L! - k! \ge L!.$$

Thus $1 < q < n - k!$, and $n - k!$ is composite.

Finally, $(n, W) = 1$ implies that n is odd. Since $n > 2L! \ge 4$, the number $n - 1$ is an even integer greater than 2. Hence $n - 1$ is composite, which settles the case $k = 1$. □

Proof of Theorem 1.2. Let $\delta \in (0, 1)$ be the constant in Theorem 1.1, and put $\eta = \delta/2$. By Propositions 2.4 and 2.5, for all sufficiently large L there is a minimum-weight rainbow edge cover $\mathcal{C}_L$ with

$$\log M(\mathcal{C}_L) = \kappa(L) \le (1 - \delta) L \log L.$$

Also, Stirling's formula gives

$$\log X \ge L \log L - 2L$$

for sufficiently large L. Therefore,

$$\begin{aligned}(1 - \eta) \log X - \log M(\mathcal{C}_L) &\ge (\delta - \eta) L \log L - 2(1 - \eta) L \\ &= \frac{\delta}{2} L \log L - 2(1 - \eta) L > 0\end{aligned}$$

once L is large enough. Thus

$$M(\mathcal{C}_L) \le X^{1 - \eta}.$$

Theorem 6.1 now gives the result after increasing the absolute threshold L_1 if necessary. □

7. Concluding remarks

The hypergraph $\mathcal{H}_L$ is not tied to gap two. By Proposition 2.3, a prime $q > L$ dividing

$$D_{i,j} = \prod_{r=i+1}^{j} r - 1$$

places i and j in a common edge. For example, the gap-three identity

$$\frac{(x+1)!}{(x-2)!} = x^3 - x$$

gives a common edge containing $x-2$ and $x+1$ whenever $q \mid x^3 - x - 1$. The corresponding auxiliary conflict graph has bounded degree when $L < q \le L^{3/2}$, so the independent-set lemma selects a fixed proportion of any available family with distinct colors and disjoint designated pairs. The unresolved arithmetic question is whether such low-weight prime divisors occur for a positive proportion of the indices.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

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